Josef Ullrich

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EDUCATION

University of California, Santa Barbara **September 2023 - June 2028**

B.S./M.S. Electrical Engineering Program (B.S. 2027, M.S. 2028)

GPA: 3.96/4.0 | Tau Beta Pi Honors Society | Engineering Honors Program

RESEARCH EXPERIENCE

Wireless Communications and Sensornets Laboratory **December 2025 - Present**

Undergraduate Researcher | Advised by Professor Upamanyu Madhow | Santa Barbara, CA

- Developed realistic user-trajectory and 3D building-occupancy mapping tools; extended CNN localization to multiple base stations by stacking beam-delay profiles as input channels.
- Ran the ray-tracing, preprocessing, training, and evaluation pipeline for 5,000 simulated positions in a four-base-station Times Square scene; achieved **3.2 m mean localization error** on a held-out validation set, with scene-geometry and base-station-placement limitations.
- Traced unrealistic building heights to Geo2SigMap filler values; integrated Overture building and building-part heights through footprint matching, reducing recognizable filler-height assignments from **22% to 0%** in a tested 200 m × 200 m scene.
- Presented the building-height investigation to Duke's **FuNCtions Laboratory** and opened a pull request at their request.
- Developed synthetic attenuation during training to address car obstructions, reducing mean localization error by **15% versus the baseline** on the car-obstruction test set.
- Project abstract accepted for the **2026 Asilomar Conference on Signals, Systems, and Computers**; coauthor on the accompanying paper, currently in preparation (final revision due December 2, 2026).

Science of Information Lab, UCSB **September 2026 - Present**

Undergraduate Researcher | Advised by Professor Qian Yu | Santa Barbara, CA

- Working on problems in coded computation and polynomial coding.

WORK EXPERIENCE

Northrop Grumman Mission Systems **June 2026 - August 2026**

Electrical Engineering Intern | San Diego, CA

- Coded, tested, and debugged **FPGA verification modules**; translated verification requirements into sequence diagrams specifying OSVVM calls and messages between components.
- Followed coding standards and documentation requirements for an Internal Critical Design Review.

Northrop Grumman Mission Systems **June 2025 - September 2025**

Systems Engineering Intern | San Diego, CA

- Supported engineering design lifecycle management through redline analysis, fit checks, and SolidWorks modeling; coordinated project blockers and action steps with engineering and manufacturing teams.

LHPH Capital **July 2024 - September 2024**

AI Implementation Intern | San Diego, CA

- Implemented an AI chatbot using proprietary datasets within Microsoft Azure's OpenAI resource and supported company-wide adoption of ChatGPT Team.

SELECTED PROJECTS

Ground-Penetrating Radar Imaging | Paper **2026**

- Reconstructed subsurface pavement images from FMCW radar data using inverse FFT processing in MATLAB; identified rebar approximately 10-15 cm below the surface.

Guitar Signal Chain and Filters | Paper, Circuits, Demonstration **2025**

- Designed, simulated in LTspice, built, and tested a guitar signal chain with buffer, preamp, and drive stages; implemented fuzz, band-pass, and delay effects and tuned component values.

Digital Highpass Butterworth Filter | Paper **2026**

- Designed and verified a digital highpass filter in MATLAB using lowpass conversion and the bilinear transformation; achieved 0.47 dB passband attenuation and 24.21 dB stopband attenuation.

SKILLS

Technical: Wireless Communications, Signal Processing, DSP, FPGA Verification, Digital System Design

Hardware & Modeling: VHDL, OSVVM, QuestaSim, LTspice, SysML

Programming & Tools: Python, MATLAB, C/C++, Linux, NVIDIA Sionna, IBM DOORS

Machine Learning: TensorFlow, Keras, CNNs, Channel Modeling